SHOPIFY SALES ANALYSIS

Business Intelligence Project Report

Submitted By

Name: **Divya Patil**

Internship Program: **Data Analyst Internship**

Organization: **Innovexis**

Internship Duration: **15 July 2026 – *15 Jan 2027***

Project Title: **Shopify Sales Analysis**

Tool Used**: Microsoft Power BI Desktop**

Dataset**: Shopify Sales Transactions Dataset**

Skills Applied:
Power Query (ETL)
Data Modeling
DAX
Data Visualization
Business Intelligence (BI)

Submitted To

Innovexis Internship Team

Submission Date

31 July 2026

INDEX

Introduction

Business Intelligence (BI) helps organizations transform raw data into meaningful insights that support informed decision-making. This project focuses on analysing Shopify sales transaction data using Microsoft Power BI to understand sales performance, customer purchasing behaviour, and customer retention. By applying Power Query for data preparation, Data Modeling for building relationships, and DAX for business calculations, the project aims to create an interactive dashboard that enables stakeholders to monitor business performance effectively.

The dashboard provides insights into key business metrics such as total revenue, total orders, average order value, repeat purchase rate, customer lifetime value, and customer segments. Interactive visualizations and filters allow users to explore the data and identify trends that can help improve sales and customer retention strategies.

In today's digital business environment, organizations generate a large volume of sales and customer data every day. However, collecting data alone is not enough; businesses need to analyse it effectively to make informed decisions. Business Intelligence (BI) plays a vital role in transforming raw data into meaningful insights through data visualization, reporting, and analytics. These insights help organizations improve operational efficiency, understand customer behaviour, identify business opportunities, and support strategic decision-making.

This project focuses on analysing **Shopify Sales** data using **Microsoft Power BI**, a leading Business Intelligence and Data Visualization tool. The Shopify dataset contains transaction-level information, including order details, customer information, product details, quantities, sales amount, taxes, and transaction dates. By analysing this data, the project aims to identify sales trends, evaluate product performance, understand customer purchasing behaviour, and measure customer retention.

The project begins with **data preparation using Power Query**, where the dataset is cleaned, validated, and transformed into a suitable format for analysis. A structured data model is then created by implementing a **Star Schema**, including a dedicated **Date Table** to support time-based analysis. Various **DAX**

(Data Analysis Expressions) measures are developed to calculate important business metrics such as Total Revenue, Total Orders, Average Order Value (AOV), Customer Lifetime Value (CLV), and Repeat Purchase Rate.

An interactive dashboard is designed to present these insights in a clear and user-friendly manner. The dashboard includes KPI cards, charts, graphs, and slicers that enable stakeholders to explore the data dynamically. Users can analyse sales performance over different time periods, identify top-performing products, compare customer segments, and evaluate long-term customer value through interactive visualizations.

The primary objective of this project is to help stakeholders make **data-driven business decisions** by providing accurate and meaningful insights into sales performance, customer behaviour, and retention. The dashboard is intended to support business users such as store owners, sales managers, marketing teams, and business analysts in monitoring key performance indicators and identifying opportunities to improve revenue growth and customer satisfaction.

Overall, this project demonstrates the practical application of **Business Intelligence concepts**, including **ETL (Extract, Transform, Load)**, **Data Modeling**, **DAX calculations**, and **Dashboard Development** using Microsoft Power BI. It showcases how raw transactional data can be transformed into actionable business insights that support strategic planning and continuous business improvement.

Stage 0 – Problem Statement

1. Project Goal

The primary goal of this project is to analyse Shopify sales data using Microsoft Power BI and develop an interactive Business Intelligence dashboard. The dashboard is designed to help stakeholders understand transaction performance, customer purchasing behaviour, and customer retention patterns through meaningful visualizations and business metrics. The insights generated from this dashboard will support data-driven decision-making and help improve overall business performance.

2. Stakeholders

The primary stakeholders for this project are:

- Store Owner
- Sales Manager
- Marketing Team
- Business Analyst
- Customer Relationship Manager (CRM)

These stakeholders can use the dashboard to monitor sales performance, identify high-value customers, evaluate product performance, and improve customer retention strategies.

3. Business Questions

The dashboard is designed to answer the following business questions:

1. What is the total revenue generated during the selected period?
2. Which products contribute the highest and lowest revenue?
3. How many customers are new compared to returning customers?
4. What is the average order value of customer purchases?
5. Which customer segments generate the highest revenue and lifetime value?

4. Key Performance Indicators (KPIs)

The following KPIs will be tracked throughout the dashboard:

- Total Revenue
- Total Orders
- Average Order Value (AOV)
- Unique Customers
- Repeat Purchase Rate
- Customer Lifetime Value (CLV)
- Revenue Growth
- Customer Retention Rate

5. Project Objective

The primary objective of this project is to build an interactive Power BI dashboard that provides comprehensive insights into Shopify sales performance and customer behaviour.

Steps to achieve the objective

1. Prepare and clean the Shopify sales dataset using Power Query.
2. Build an optimized data model and create business measures using DAX.
3. Design an interactive dashboard with business insights and recommendations.

Stage 1 – Data Preparation & Modeling

1.1 Data Import

The Shopify Sales dataset was imported into Microsoft Power BI Desktop using the Get Data feature. The dataset contains 7,431 transaction records stored in a single fact table named shopify_sales.

The dataset includes important business fields such as:

Column	Description
Order Number	Unique identifier for each order
Customer ID	Unique customer identifier
Invoice Date	Date and time of the transaction
Product ID	Product identifier
Product Type	Product category
Quantity	Quantity purchased
Subtotal	Product amount before tax
Total Sales	Final sales amount
Total Tax	Tax amount collected
City	Customer location

1.2 Data Cleaning using Power Query

Power Query was used to prepare the dataset before analysis. The following data cleaning operations were performed:

Cleaning Step	Purpose
Verified column data types	Ensured accurate calculations and analysis
Checked missing values	Identified null or blank records
Validated duplicate records	Ensured data consistency
Applied Trim	Removed leading and trailing spaces from text fields
Applied Clean	Removed hidden and non-printable characters
Renamed columns	Improved readability and consistency
Created helper columns	Supported time-based analysis and reporting

1.3 Helper Columns Created

The following helper columns were created in Power Query:

Helper Column	Purpose
Order Year	Analyse yearly sales
Order Month	Monthly analysis
Month Name	Improve chart readability
Order Quarter	Quarterly performance analysis
Day Name	Analyse sales by day of the week

1.4 Data Model

A Star Schema was implemented for the project.

The shopify_sales table was used as the Fact Table, while a separate Date Table was created as the Dimension Table.

A one-to-many relationship was established between:

Date Table[Date] → shopify_sales[Invoice Date]

This relationship enables accurate filtering, time-based analysis, and Power BI time intelligence functions.

1.5 Date Table

A dedicated Date Table was created using the following DAX formula:

```
Date Table =
CALENDAR(
   MIN('shopify_sales'[Invoice Date]),
   MAX('shopify_sales'[Invoice Date])
)
```

Explanation: This DAX formula generates a continuous calendar between the earliest and latest transaction dates in the dataset.

The Date Table was connected to the shopify_sales table and recognized by Power BI for time-based analysis.

STAGE 2 – Transaction Performance Analysis

2.1 Create DAX Measures for Total Revenue, Total Orders, and Average Order Value (AOV). Display them as KPI Cards.

To understand the overall sales performance, three important DAX measures were created in Power BI. These measures help summarize the sales data into meaningful business metrics.

Total Revenue

DAX Formula

Total Revenue =
SUM(shopify_sales[Total Sales])

Explanation:
This measure calculates the total sales revenue generated from all customer orders. It shows the overall income earned by the business.

Total Orders

DAX Formula

Total Orders =
DISTINCTCOUNT(shopify_sales[Order Number])

Explanation:
This measure counts the total number of unique orders placed by customers. Since one order can contain multiple products, `DISTINCTCOUNT()` ensures that each order is counted only once.

Average Order Value (AOV)

DAX Formula

Average Order Value =
DIVIDE([Total Revenue],[Total Orders],0)

Explanation:
This measure calculates the average amount spent by a customer in a single order. It is useful for understanding customer purchasing behaviour.

KPI Cards

The above measures were displayed as KPI Cards on the dashboard.

Purpose:
KPI Cards provide a quick summary of the business by displaying Total Revenue, Total Orders, Average Order Value, and Unique Customers. This allows stakeholders to understand the overall business performance at a glance.

2.2 Visualise Revenue Over Time (By Month/Quarter/Year)

A Line Chart was created using the Date field on the X-axis and Total Revenue on the Y-axis.

Explanation:
This chart helps analyse how revenue changes over time. It makes it easy to identify whether sales are increasing, decreasing, or remaining stable during the available period.

Observation:
The dataset contains sales data for March 2025. The chart shows daily changes in revenue during this period. Since the dataset covers only a short time, long-term seasonal trends cannot be identified.

2.3 Revenue by Product Category

A Clustered Bar Chart was created using Product Type and Total Revenue.

Explanation:
This chart compares the revenue generated by different product categories. It helps identify which products contribute the most to total sales and which products generate lower revenue.

Observation:
The chart provides a clear comparison of product performance, helping the business identify top-selling and low-performing products for better inventory and marketing decisions.

2.4 Analyse Orders by Day of the Week

A Clustered Column Chart was created using Day Name and Total Orders.

Explanation:
This chart shows the number of customer orders placed on each day of the week. It helps identify the busiest and least busy shopping days.

Observation:
By understanding customer purchasing patterns, the business can plan promotions, manage inventory, and schedule staff more effectively on high-demand days.

2.5 Revenue Growth Compared with the Previous Period

To analyse business growth, two DAX measures were created: Revenue Previous Period and Revenue Growth %.

A Line Chart was used to display Revenue Growth over time.

Explanation:
This analysis compares the current revenue with the previous period to understand whether sales are increasing or decreasing.

Observation:
The chart helps track short-term business performance and shows changes in revenue over the available period. It can be used to monitor sales trends and evaluate business growth.

STAGE 3 – Customer Purchasing Behaviour

Objective

The objective of this stage was to analyse customer purchasing behaviour by identifying unique, new, and returning customers, understanding purchase frequency, recognising high-value customers, segmenting customers based on their spending patterns, and analysing the popularity of different product categories. This analysis helps businesses understand customer engagement, improve retention strategies, and focus marketing efforts on valuable customer segments.

1. Customer Analysis

To understand the customer base, three DAX measures were created:

- **Unique Customers** – Calculates the total number of distinct customers.
- **New Customers** – Calculates customers who have placed only one order.
- **Repeat Customers** – Calculates customers who have placed more than one order.

These measures were displayed using KPI cards to provide a quick overview of customer distribution.

Observation

The dashboard shows that the majority of customers are first-time buyers, while a smaller percentage consists of repeat customers. Increasing repeat purchases can improve customer retention and long-term business growth.

2. Order Frequency Analysis

A separate **Customer Order Frequency** table was created to calculate the number of orders placed by each customer.

This analysis helps identify how frequently customers purchase products from the store.

Observation

The analysis indicates that most customers placed only one order, while comparatively fewer customers made multiple purchases. This suggests that customer retention strategies such as loyalty programs, personalised offers, and follow-up marketing campaigns could encourage repeat purchases.

3. Top Customers by Total Spend

A clustered bar chart was created to identify the top customers based on their total spending.

The visual highlights the customers who contribute the highest revenue to the business.

Observation

A small group of customers contributes significantly to the overall revenue. Identifying these high-value customers allows businesses to focus on customer loyalty programs, exclusive offers, and personalised marketing strategies to maintain long-term relationships.

4. Customer Segmentation

Customers were segmented based on their **total spending** using a calculated table (**Customer Spend**) and a calculated column (**Customer Segment**).

The customers were classified into three categories:

- High Value Customers
- Medium Value Customers
- Low Value Customers

A donut chart was created to visualise the distribution of customers across these segments.

Observation

The segmentation helps identify which customers generate the highest revenue. Businesses can prioritise high-value customers with premium services while designing promotional campaigns to encourage medium- and low-value customers to increase their purchases.

5. Product Type Analysis

A clustered bar chart was created using **Product Type** and **Total Revenue** to identify the most popular product categories.

Observation

The analysis highlights which product categories generate the highest revenue. This information helps businesses improve inventory planning, focus marketing efforts on high-performing products, and recommend popular products to customers.

Stage 3 Summary

Stage 3 focused on understanding customer purchasing behaviour through customer analysis, purchase frequency, top customer identification, customer segmentation, and product category performance. The insights obtained from this stage help businesses improve customer retention, personalise marketing strategies, and maximise revenue by focusing on valuable customers and popular products.

DAX Measures Used in Stage 3

Measure / Table	Purpose
Unique Customers	Counts the total number of distinct customers.
Repeat Customers	Counts customers who placed more than one order.
New Customers	Calculates customers who placed only one order.
Customer Order Frequency	Calculates the number of orders placed by each customer.
Customer Spend	Calculates the total spending for each customer.
Customer Segment	Categorises customers into High, Medium, and Low Value based on total spending.

STAGE 4 – Retention & Long-Term Customer Value

1. Customer Retention

Explanation

Customer retention refers to the ability of a business to retain its customers by encouraging repeat purchases. The **Repeat Purchase Rate** was calculated to determine the percentage of customers who placed more than one order.

Observation

The Repeat Purchase Rate indicates the proportion of customers who returned to make additional purchases. Customers who make repeat purchases contribute to stable revenue and demonstrate customer loyalty. Improving customer retention can significantly increase long-term business growth.

2. Cohort Analysis

Explanation

A simplified cohort analysis was performed by grouping customers according to the month of their first purchase. This helps identify customer acquisition periods and serves as a foundation for analysing customer retention.

Observation

The available dataset contains transactions only for **March 2025**, resulting in a single customer cohort. Due to the absence of data for subsequent months, long-term retention across multiple months could not be analysed. With additional historical data, the business could monitor customer retention trends more effectively.

3. Customer Lifetime Value (CLV)

Explanation

Customer Lifetime Value (CLV) was estimated using the formula:

CLV = Average Order Value × Purchase Frequency × Customer Lifespan

Since the dataset contains only one month of data, the customer lifespan was estimated as one month.

Observation

The analysis indicates that **High Value Customers** contribute the highest Customer Lifetime Value. These customers generate a significant share of the company's revenue and should be prioritised through loyalty programs, personalised offers, and excellent customer service.

4. Gap Between First and Second Purchase

Explanation

The analysis aimed to determine the average time gap between a customer's first and second purchase.

Observation

As the dataset contains transactions for only one month, it was not possible to calculate the average time gap between the first and second purchase accurately. With data covering multiple months, the business could identify the ideal time to send promotional offers and re-engagement campaigns.

5. Customer Retention Strategy

Explanation

Customer retention strategies were developed based on Repeat Purchase Rate and Customer Lifetime Value.

Observation

The business should focus on retaining **High Value Customers**, as they contribute the highest revenue. Encouraging first-time customers to make repeat purchases through loyalty programs, personalised recommendations, and promotional campaigns can improve long-term profitability.

STAGE 5 – Dashboard Design & Insights

1. Dashboard Design

Explanation

The dashboard was designed as an interactive business intelligence solution consisting of three pages. The first page presents key performance indicators (KPIs), the second page focuses on transaction performance analysis, and the third page provides customer purchasing behaviour and retention insights.

Observation

The dashboard provides a structured and user-friendly layout that allows stakeholders to quickly understand overall business performance and drill down into detailed sales and customer insights.

2. Dashboard Layout

Explanation

The dashboard follows a logical structure, beginning with high-level KPIs, followed by sales trends and ending with customer behaviour analysis.

Observation

This layout improves readability and enables decision-makers to move from summary information to detailed business analysis without confusion.

3. Dashboard Interactivity

Explanation

Interactive slicers and cross-filtering features were implemented to enhance dashboard usability.

Observation

Users can filter the dashboard using Invoice Date and Product Type, allowing them to explore business performance dynamically. Selecting one visual automatically filters related visuals, making the report interactive and easy to analyse.

4. Key Insights

Insight 1

The business generated consistent revenue during the available reporting period, with sales concentrated in March 2025.

Insight 2

Most customers placed only one order, indicating an opportunity to improve repeat purchases and customer retention.

Insight 3

A small group of customers contributed a significant portion of total revenue, highlighting the importance of retaining high-value customers.

Insight 4

High Value Customers generated the highest Customer Lifetime Value and should be prioritised through personalised engagement strategies.

Insight 5

Certain product types generated higher revenue than others, indicating opportunities to focus inventory management and marketing efforts on top-performing products.

5. Business Recommendations

- Introduce loyalty and rewards programs to encourage repeat purchases.
- Offer personalised product recommendations based on customer purchase history.
- Reward high-value customers with exclusive discounts and early access to new products.
- Focus promotional campaigns on the highest-performing product categories.

- Continue monitoring customer behaviour using Power BI dashboards to support data-driven decision-making.

Final Conclusion

This project successfully analysed Shopify sales data using Microsoft Power BI. Data cleaning, transformation, DAX calculations, and interactive visualisations were used to evaluate sales performance, customer purchasing behaviour, and customer retention. The dashboard provides valuable insights into revenue trends, product performance, customer segments, and long-term customer value. The findings and recommendations can help businesses improve customer retention, increase revenue, and support informed business decisions through data-driven analytics.

Conclusion

This project successfully analysed Shopify sales data using Microsoft Power BI to gain meaningful business insights. The data was cleaned and transformed using Power Query, and DAX measures were created to calculate important business metrics such as Total Revenue, Total Orders, Average Order Value (AOV), Customer Lifetime Value (CLV), and Repeat Purchase Rate.

Interactive dashboards were developed to analyse transaction performance, customer purchasing behaviour, and customer retention. The analysis identified high-performing products, valuable customer segments, and opportunities to improve customer retention through data-driven strategies. Although the dataset was limited to one month, the project demonstrated how business intelligence tools can support decision-making by presenting information through clear visualisations and KPI dashboards.

Overall, this project enhanced practical skills in data preparation, data modelling, DAX, dashboard design, and business analytics using Power BI. The insights and recommendations generated from this analysis can help businesses improve operational performance, customer satisfaction, and long-term revenue growth.

References

The following resources were referred to while completing this project:

1. **Microsoft Power BI Documentation**
   https://learn.microsoft.com/power-bi/
2. **Microsoft Learn – Power BI Learning Path**
   https://learn.microsoft.com/training/powerplatform/power-bi/
3. **Microsoft DAX Documentation**
   https://learn.microsoft.com/dax/
4. **Shopify Documentation**
   https://shopify.dev/docs
5. **Power BI Community**
   https://community.powerbi.com/
6. **Microsoft Power Query Documentation**
   https://learn.microsoft.com/power-query/
7. **Course Material and Internship Guidelines Provided by Innovexis**

THANK YOU